PUSHKARAJ BARADKAR

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EDUCATION

University of Southern California, Viterbi School of Engineering GPA: 3.73 / 4.00 — **Jan 2025 - Dec 2026**
Master of Science in Computer Science — Los Angeles, CA
Machine Learning for Data Science, Analysis of Algorithms, Web Technologies, Database Systems

University of Mumbai GPA: 9.13 / 10 — **Jul 2020 - May 2024**
Bachelor of Engineering in Computer Engineering — Mumbai, India
Data Structures, Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Natural Language Processing

EXPERIENCE

AI Engineer Intern | Tabhi, Austin, TX — **May 2026 – Present**

- Built a TurboVec index retrieval + Two-Tower recommendation system serving 300K+ users with less than 100ms latency.
- Architected a Two-Tower ranking pipeline using 100+ behavioral and graph features, improving Recall@100 and NDCG@10 by ~18%.
- Implemented evaluation frameworks using Recall@K, NDCG, and MRR to benchmark retrieval and ranking models.
- Automated evaluation and benchmarking across 10+ ranking and grading metrics, accelerating experimentation by ~3×.
- Engineered CLIP-based embeddings for places, restaurants, and experiences, supporting retrieval across 500K+ items.

Software Engineer Intern – Real-Time AI Systems | USC, Los Angeles, CA — **Nov 2025 – May 2026**

- Built a real-time SLAM pipeline combining monocular VO and depth estimation, achieving ~20–25 FPS on noisy indoor data.
- Implemented RANSAC-based feature matching (2500 matches, 2000 inliers), leveraging IMU to improve motion consistency under noise.
- Resolved scale ambiguity using sparse depth and fused VO with GPS in GTSAM, reducing trajectory error from ~95 m to ~1.2 m ATE.
- Debugged sensor noise, calibration drift, and VO failures, improving tracking stability and reducing dropouts by ~30%.

Software Engineer Intern – AI & Computer Vision | YMT Medical, India — **Jan 2024 – Aug 2024**

- Engineered computer vision pipelines for image quality enhancement and denoising, improving robustness of AI-driven diagnostics.
- Trained and fine-tuned YOLOv8n on 2K dermatology images, supporting model validation and performance testing.
- Enhanced GPU inference and integrated clinical feedback to enhance model interpretability and reduce latency by 40%.
- Designed evaluation pipelines to measure model performance, interpretability, and latency tradeoffs in production-like settings.

Software Engineer Intern – Backend & APIs | Technoriya ERP Solution, India — **Oct 2023 – Dec 2023**

- Developed and optimized backend services and APIs using Python and Node.js, supporting scalable data pipelines, system reliability, cloud-based deployment and processing layers for downstream analytics and ML workflows.
- Implemented secure authentication and access protocols via Firebase Auth and Firestore rules to ensure data integrity.
- Optimized RESTful APIs and Firestore queries to achieve 450 ms response times under 1,000 concurrent requests for ML integration.
- Streamlined deployment workflows using Vercel and GitHub Actions to improve delivery consistency and reliability.

PROJECTS

Godot MCP: AI-Native Game Development Interface | TypeScript, Node.js, MCP - Link — **April 2026 – Present**

- Built an MCP server enabling LLM agents (Claude, Cursor) to control Godot projects via natural language; used by 10+ active users.
- Developed 30+ tools for scene, script, execution, and asset management with secure path validation and structured routing.
- Implemented CI/CD, auto-config detection, and multi-client integration for end-to-end 2D/3D workflow automation.

AgentPay: Autonomous Agent Economy | Solana, Avalanche, LangChain, Next.js - Link — **April 2026 - Present**

- Built a multi-agent system where AI agents autonomously discover, hire, and pay other agents using on-chain payments and reputation.
- Designed smart contract system (Rust/Anchor) for agent registry, escrow, and staking/slashing with verifiable reputation tracking.
- Integrated LangChain-based orchestration with custom payment tooling (x402), enabling under 30s end-to-end agent interaction cycles.

Open Source Contribution – Defillama SDK (RPC Reliability & Failover) | TypeScript, Node.js, Web3 - Link — **April 2026**

- Fixed RPC override bug (Fixes #162) by redesigning Chain RPC logic to correctly prioritize user-defined endpoints.
- Implemented runtime RPC quarantine (5-failure threshold, 10 min cooldown), reducing repeated retries and improving request reliability.
- Added 8 unit tests covering override and failure-handling logic; expanded SDK documentation for RPC configuration.

TECHNICAL SKILLS, PUBLICATIONS, CONFERENCES

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, R, C#
- **AI / ML & Data Science:** Deep Learning, Generative AI, NLP, LLMs (ChatGPT, Claude Code, Copilot), Transformers, RAG, Agentic AI Systems, Autonomous Agents, Prompt Engineering, Computer Vision, Data Mining, Data Science, PyTorch, TensorFlow, LangChain, Pandas, NumPy, Matplotlib, LangGraph, MCP, Google ADK(Agent Development Kit), A2A
- **Frameworks & Development:** FastAPI, Flask, Django, Node.js, Express.js, React.js, Next.js, Angular.js, OAuth, OOP, Agile, HTML/CSS
- **Databases & Cloud:** MongoDB, SQL, PostgreSQL, Supabase, Redis, AWS, GCP, Azure, Docker, Kubernetes, Git, CI/CD, Gitlab, Linux
- **Software Engineering:** Object-Oriented Programming (OOP), design patterns, backend development, REST APIs, debugging, Agile, REST API, AWS, Lambda, EC2, Fargate, CloudWatch, SQS, SNS, SES, SageMaker